

# Week 4 Notes

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## 1 Week 4 Notes

### 1.1 What is a function?

Simply, a function is a grouping of code that can be reused. A function is defined by first using the keyword **function**, and it is *used* by **calling** the function.

#### 1.1.1 Syntax Example

```
// function declaration
function foo() {
    document.writeln("bar");
}

// function call
foo();
```

### 1.1.2 Anatomy of a function

**Function** Keyword that starts the function.

**Function Name** The function name follows the function keyword.

**The Parentheses** The parentheses are what hold the function parameters.

**Curly Braces** The curly braces hold the block of code that will be executed when the function is called.

**The function call** Technically, the function call isn't part of the function, though it's necessary for the function to actually do something. If you want to use the function later, you must *call* it.

## 1.2 Constants

Unlike C/C++ `const` is a little less strict. The only similarity to C++ is that a `const` in JavaScript must be defined on declaration.

### 1.2.1 Syntax

```
const FOO = "bar";
```

### 1.2.2 Rules for ITPJS:

- A constant must be declared for at the top of the list, above the `let` variables.
- A constant must be named in all uppercase. If it's two words, separate by *underscores*.
- When you have a value that you want to remain *constant* or immutable, use `const` instead of `let`.

## 1.3 Addition Operator

In JS and many other programming languages, the `+`operator is used for adding two operands together, but it's also used for **string concatenation**.

### 1.3.1 Addition vs Concatenation Rules

= If the + has a **string** on either side of it, it will convert the number to a **string** and concatenate.

```
// Additon  
1 + 2;
```

```
// String Concatenation  
"string1" + "string2";
```